Skills Summary

Rational and Irrational Numbers

Use this reference while completing your worksheet or when studying.

One definition, one test. A number is **rational** when it can be written as $\dfrac{p}{q}$ with p, q integers and $q \neq 0$; otherwise it is **irrational**.

Always rational	integers, fractions, terminating decimals, repeating decimals, roots of perfect squares
Irrational	π, and $\sqrt{n}$ whenever n is a whole number that is not a perfect square

The decimal is the giveaway: *ends* or *repeats* $\rightarrow$ rational; *never ends and never repeats* $\rightarrow$ irrational.

1. Roots of perfect squares (perfect-square-root)

$1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196$ are the perfect squares up to 14^2.
If the radicand is in that list, the root is a whole number, so $\sqrt{n} = \frac{\text{whole number}}{1}$ is **rational**.
The same works for fractions: $\sqrt{p/q} = \sqrt{p}/\sqrt{q}$.

Example: Classify $\sqrt{81}$.
Step 1. The question is whether the root is exact, so check the radicand against the perfect-square list rather than reaching for a calculator.
Step 2. 81 is in the list, so $\sqrt{81} = 9 = \frac{9}{1}$ — a ratio of integers. $\Rightarrow$ **9**, and $\sqrt{81}$ is **rational**.

Try it: Classify $\sqrt{144}$ and give its exact value.

check: rational, **12**

2. Roots that are not exact (nonperfect-root)

If the radicand sits *between* two perfect squares, the root sits between their roots and the decimal never ends or repeats — it is **irrational**. Bracket it first, then round:

$$p^2 < n < q^2 \quad \Longrightarrow \quad p < \sqrt{n} < q$$

A rounded decimal is a *stand-in*, never an equality.

Example: Locate and classify $\sqrt{20}$.
Step 1. Since 20 is not a perfect square, the useful move is to trap it between the two nearest ones instead of hunting for an exact value.
Step 2. $16 < 20 < 25$, so $4 < \sqrt{20} < 5$; a calculator gives $4.4721\ldots$, which rounds to **4.47**.
$\Rightarrow$ $\sqrt{20}$ is **irrational**, and it sits just under the midpoint of 4 and 5.

Try it: Between which two whole numbers is $\sqrt{40}$, and what is it to two decimal places?

check: between 6 and 7, **6.32**

3. Decimals and fractions (decimal-and-fraction)

Terminating: read the decimal aloud and write what you hear, then reduce. $0.7 = \frac{7}{10}$, $0.29 = \frac{29}{100}$, $0.413 = \frac{413}{1000}$.
Repeating: put the repeating block over as many 9s as it has digits. $0.\overline{4} = \frac{4}{9}$, $0.\overline{45} = \frac{45}{99}$.
Either way you end with a fraction, so both kinds are **rational**.

Example: Show that 0.125 is rational.
Step 1. It terminates, so the definition can be met directly — name the place value and write the fraction it says.
Step 2. "One hundred twenty-five thousandths" is $\dfrac{125}{1000}$, and dividing top and bottom by 125 gives the lowest-terms form. $\Rightarrow \quad 0.125 = \mathbf{\frac{1}{8}}$, so it is **rational**.

Try it: Write 0.24 as a fraction in lowest terms.

check: $\mathbf{\frac{6}{25}}$

4. Comparing and ordering (compare-order)

Put every number in the *same* form — decimals are easiest. Fractions: divide. Roots: bracket with perfect squares, then refine by squaring your guesses. Size, not notation, decides the order.

Example: Order 2.5, $\sqrt{7}$, $\frac{11}{4}$ from least to greatest.
Step 1. The three are written in three different notations, so convert to a common decimal form before comparing anything.
Step 2. $\frac{11}{4} = 2.75$; and $4 < 7 < 9$ gives $2 < \sqrt{7} < 3$, with $2.75^2 = 7.5625 > 7$, so $\sqrt{7} < 2.75$.
$\Rightarrow \quad \mathbf{2.5 < \sqrt{7} < \frac{11}{4}}$

Try it: Order 1.7, $\sqrt{3}$, $\frac{9}{5}$ from least to greatest.

check: $\mathbf{1.7 < \sqrt{3} < \frac{9}{5}}$

(!) Watch out: a radical sign does not make a number irrational ($\sqrt{9}$ is just 3), and a calculator display is not proof of anything — it truncates every irrational number to a terminating, and therefore rational, decimal.